Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, decision-making, language understanding, and image recognition.

Key Features of AI

- Learning from data
- Recognizing patterns
- Understanding human language
- Making predictions and decisions
- Automating repetitive tasks

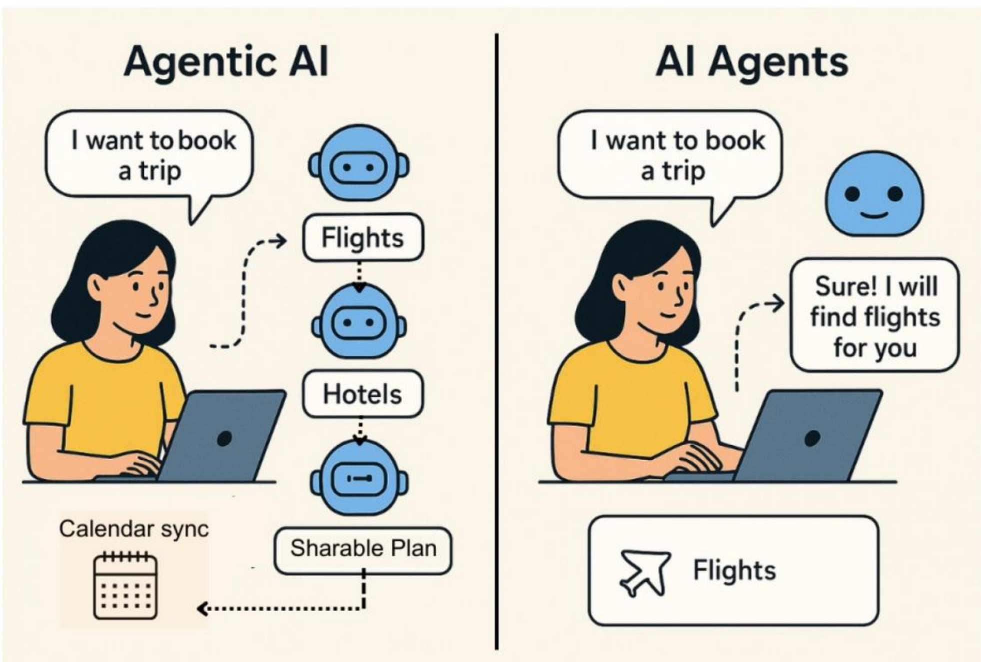
Applications of AI

- Virtual assistants (ChatGPT, Siri, Alexa)
- Self-driving vehicles
- Medical diagnosis
- Recommendation systems

Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Technology	Definition	Example
Artificial	Broad field of creating intelligent machines.	Self-driving cars

Technology	Definition	Example
Intelligence (AI)		
Machine Learning (ML)	Subset of AI where systems learn from data without explicit programming.	Spam email detection
Generative AI	AI that creates new content such as text, images, audio, or code.	ChatGPT, image generators
Agentic AI	AI systems that can plan, make decisions, and perform tasks autonomously.	AI assistant that books meetings and sends emails



Three Open-Source AI Models Available on Hugging Face

1. Llama 3

- Developed by Meta
- Open-source large language model
- Used for chatbots, content generation, and coding assistance
- Supports multiple languages

2. Mistral 7B

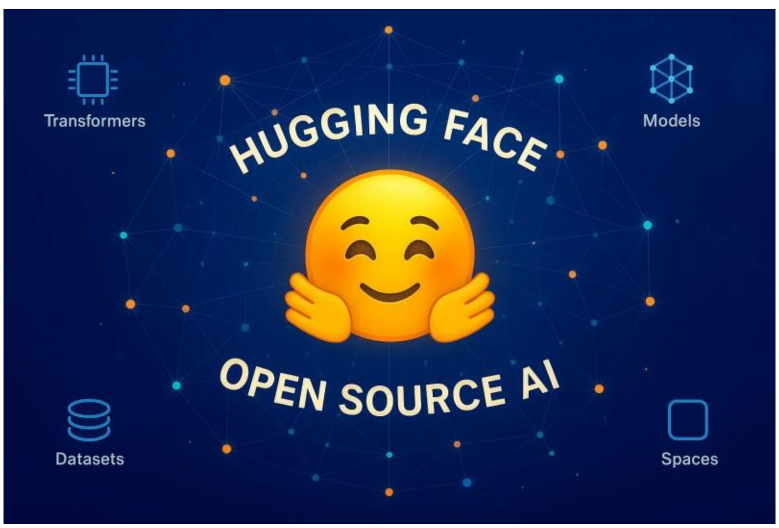
- Developed by Mistral AI
- Lightweight and efficient
- Requires less computing power
- Suitable for local deployment

3. FLAN-T5

- Developed by Google
- Fine-tuned for following instructions
- Performs summarization, translation, and question answering
- Popular for research and education

Common Uses

- Chatbots
- Text summarization
- Translation
- Code generation
- Research projects



Difference Between Traditional Programming and AI-Assisted Development

Traditional Programming	AI-Assisted Development
Rules are explicitly coded by programmers.	AI helps generate code and solutions.
Fixed logic and outputs.	Learns patterns from data and suggestions.

Traditional Programming	AI-Assisted Development
Time-consuming development.	Faster development process.
Manual debugging and coding.	AI assists with debugging and code completion.
Less adaptive.	Can adapt based on training data and context.

AI Hallucinations

AI hallucinations occur when an AI system generates information that sounds correct and confident but is actually false, misleading, or unsupported by facts.

Example

Question: Who invented the first smartphone in 1900?

Hallucinated Answer:

"John Smith invented the first smartphone in 1900."

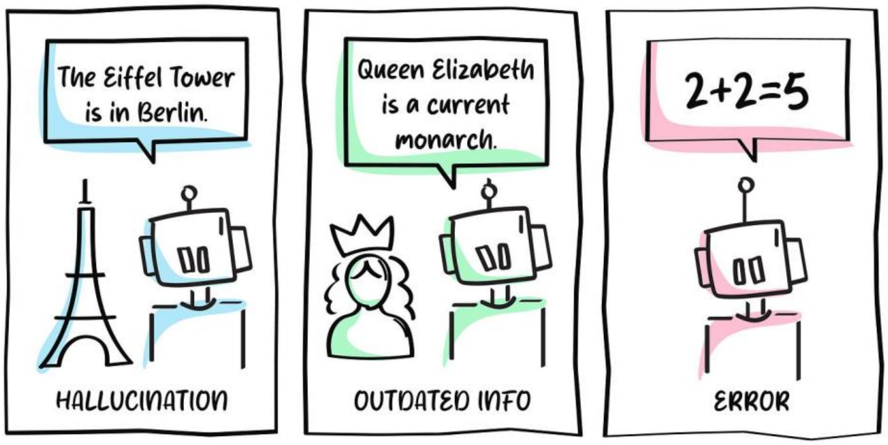
This answer sounds believable, but it is incorrect because smartphones did not exist in 1900.

Why Hallucinations Occur

- Insufficient training data
- Ambiguous questions
- Lack of real-time verification
- Statistical prediction errors

How to Reduce Hallucinations

- Verify information from reliable sources
- Use fact-checking tools
- Provide clear prompts
- Use retrieval-based AI systems


The Eiffel Tower is in Berlin.
HALLUCINATION
Queen Elizabeth is a current monarch.
OUTDATED INFO
2+2=5
ERROR